CNN-RNN Hybrid Model for Intracranial Hemorrhage Detection and Subtype Classification

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Introduction

- **Intracranial hemorrhage (ICH)** is a life-threatening condition caused by **bleeding within the skull**.
- It can result from trauma, hypertension or vascular abnormalities, requiring **rapid and accurate diagnosis**.
- Different subtypes epidural, subdural, subarachnoid, intraventricular and intraparenchymal need distinct treatment strategies.
- Traditional diagnosis relies on expert radiologists interpreting CT scans, which can be **time-consuming and error-prone**.
- This project explores how well deep learning models, trained on a subset of the RSNA ICH dataset [1], can perform **multi-label classification** of ICH subtypes.

Data Pre-Processing

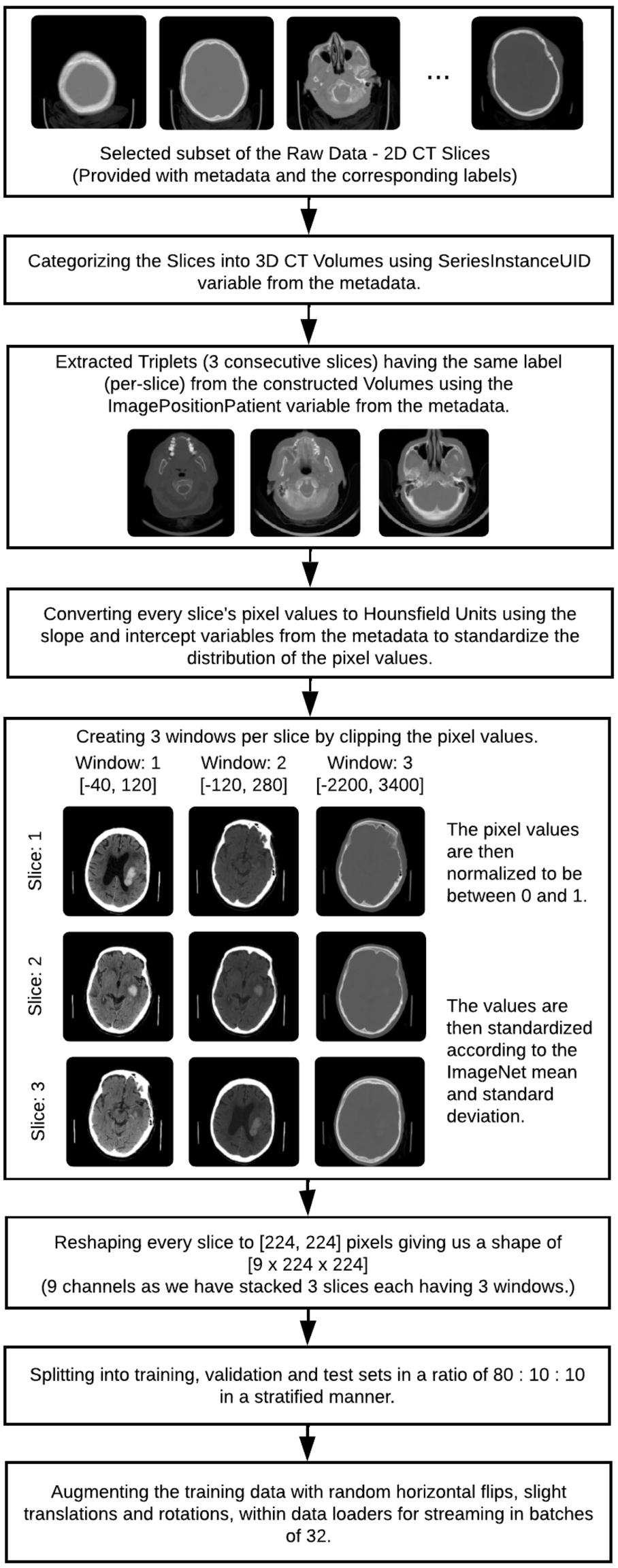


Figure 1: Overview of the data preprocessing pipeline.

Model Architecture

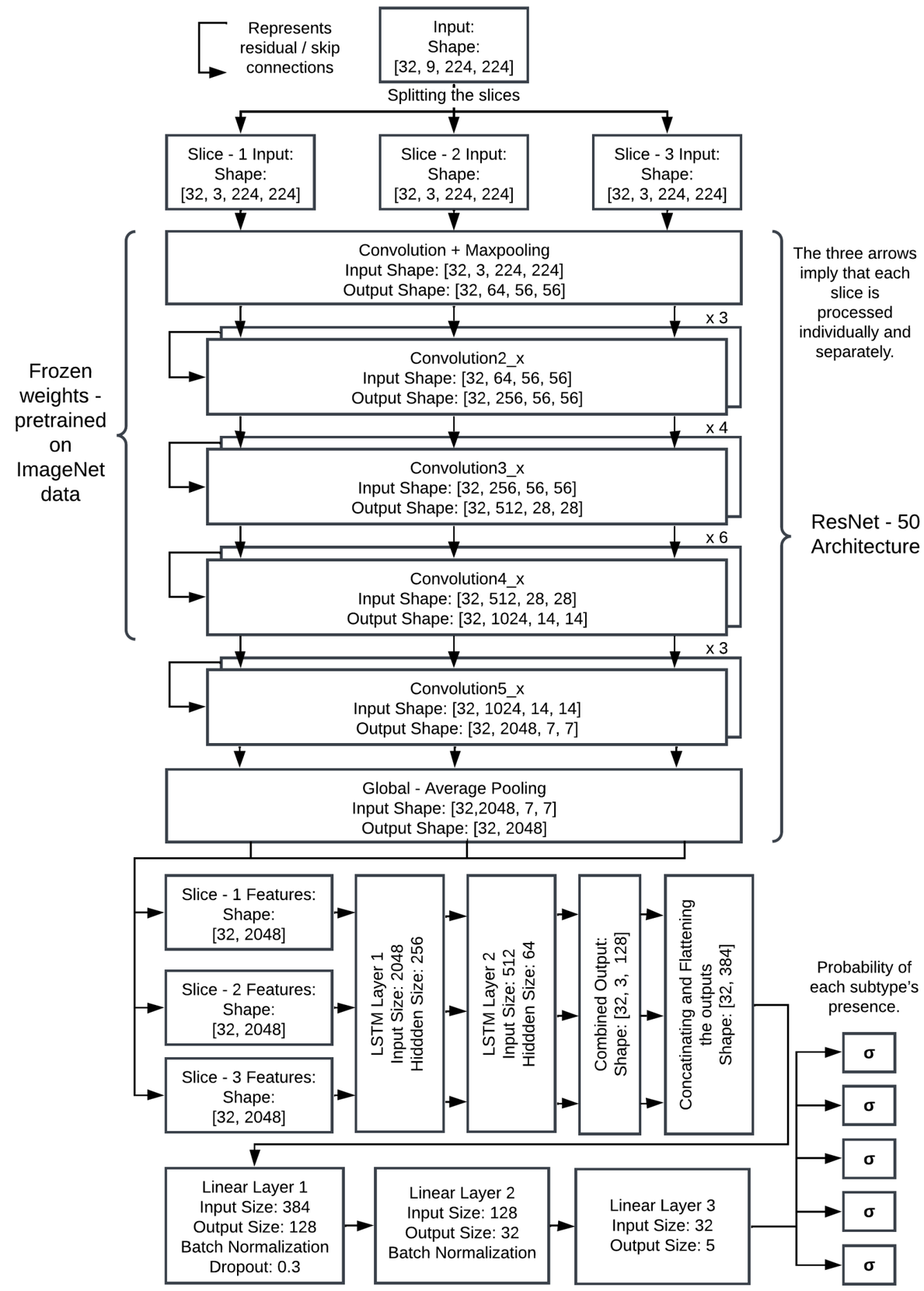
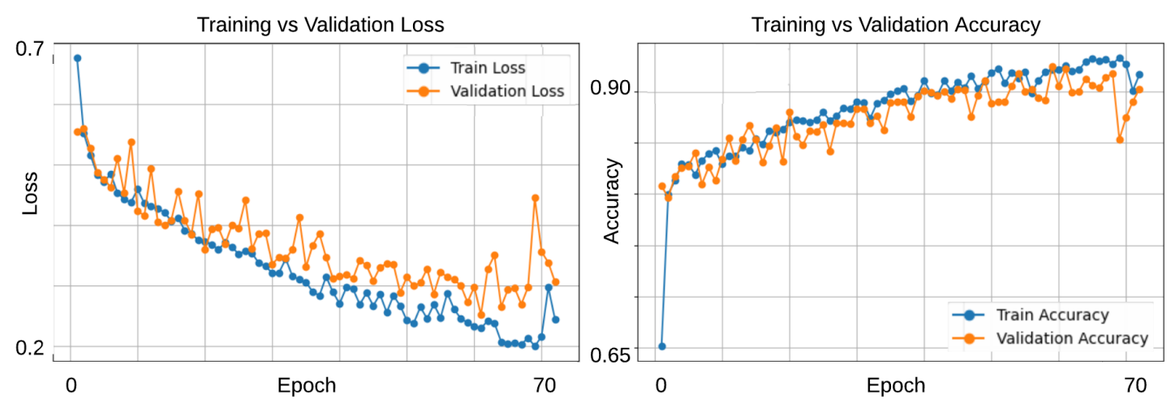


Figure 2: Overview of the proposed ResNet–LSTM model architecture.

Training and Parameters



- Trained with Adam optimizer (lr = 1×10^{-3}, weight decay = 1×10^{-6}).
- Used BCEWithLogitsLoss with class imbalance handled via positive class weights: $w_i = \log(1+N_{pos,i} / N_{neg,i})$. where $N_{pos,i}$ and $N_{neg,i}$ are the the number of positive and negative samples for class i, respectively.
- Early stopping (patience = 10, max = 100 epochs) to prevent overfitting and enhance generalization.

Figure 3: Loss and accuracy during training, early stopping at 73 epochs

Results

Table 1: Per-class performance metrics for the five intracranial hemorrhage subtypes on the test set.

Metric	Epidural	Subdural	Subarachnoid	Intraventricular	Intraparenchymal
Accuracy	0.948	0.930	0.918	0.884	0.884
Sensitivity	0.727	0.857	0.923	0.667	0.787
Specificity	0.959	0.954	0.915	0.969	0.917
Precision	0.471	0.857	0.762	0.898	0.774
NPV	0.986	0.954	0.976	0.878	0.922
ROC AUC	0.94	0.98	0.97	0.94	0.92

[1] A. E. Flanders, L. M. Prevedello, G. Shih, S. S. Halabi, J. Kalpathy-Cramer, R. Ball, J. T. Mongan, A. Stein, and K. et.al. Construction of a Machine Learning Dataset through Collaboration: The RSNA 2019 Brain CT Hemorrhage Challenge. *Radiology: Artificial Intelligence*, 2(3):e190211, May 2020.

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